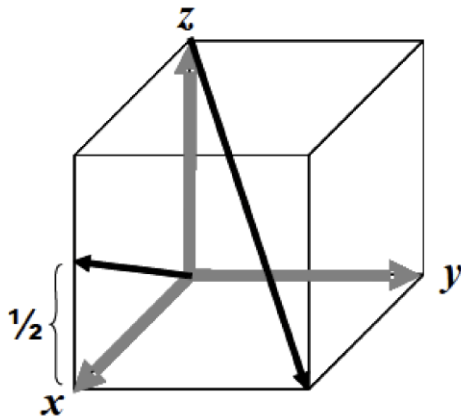
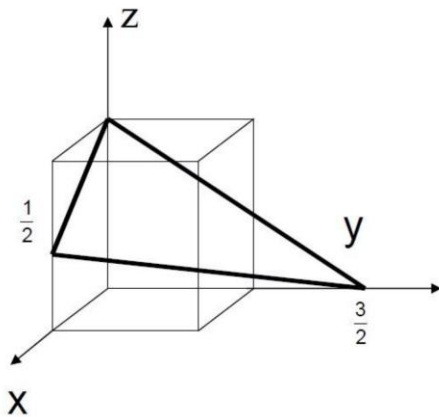


Q1) Miller index of a direction (10 pts)

Determine the Miller indices of the two directions in the figure below.

**Q2) Miller index of a plane (10 pts)**

Determine the Miller indices of the plane in the figure below.

**Q3) We are family. (15 pts)**

- How many directions belong to the family of direction $\langle 110 \rangle$ in a cubic crystal system? Explain your answer.
- How many directions belong to the family of direction $\langle 110 \rangle$ in a tetragonal crystal system? Explain your answer.
- Is the number of planes belonging to the $\{110\}$ family the same or different in tetragonal and orthorhombic crystal systems? Explain your answer.

Q4) 3D Bravais lattice: (10 pts)

Draw a tetragonal P lattice; place an extra lattice point in $(\frac{1}{2}, \frac{1}{2}, 0)$, i.e., make a C centering of the lattice. Now, show that a tetragonal C lattice does not exist, i.e. that it is equivalent to (another) P lattice.

Q5) Crystal structure visualization: (55 pts)

The unit cell of an oxide crystal with a cubic lattice parameter $a = 4.00 \text{ \AA}$ contains the following atoms (the ordered sets are the coefficients of location that define r_j):

A at (0,0,0); B at $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$;

O at $(\frac{1}{2}, \frac{1}{2}, 0)$; O at $(0, \frac{1}{2}, \frac{1}{2})$; O at $(\frac{1}{2}, 0, \frac{1}{2})$.

- Specify the Bravais lattice of this crystal.
- Specify the basis of this crystal.
- Specify the formula unit of this compound.
- Sketch the unit cell (perspective view), including all atoms that are within or partially within the cell, making sure to replicate atoms at corners, edges and facets, as needed. (hint: use different colors or patterns to identify atoms)
- What are the coordination numbers of A and B?
- Determine the A-O and B-O bond lengths in $\text{\AA}$.
- If A and B are not the same size, which is smaller? Explain your rationale?
- Specify the sequence of atoms found along the $\langle 111 \rangle$ and $\langle 110 \rangle$ directions.
- Draw the contents of the (200) plane.
- Draw the contents of the (110) plane.